An information-content ceiling check for the CNP-AIRT ablation ladder

Working note — not part of the NeurIPS submission

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1 Purpose

Every neural variant in the ablation ladder (`ablation-results.md`, `uniform-audit.md`) ties or loses to the Beta-Binomial baseline on calibrated Brier ALC. Before attributing this to undertraining or an objective bug, check whether the baseline is already near the *information-theoretic ceiling* for this task: with at most 31 binary responses, there is a hard limit on how precisely a scalar ability θ can be known, and therefore a hard floor on achievable Brier risk. If the Beta-Binomial baseline already sits at that floor, no architecture change can beat it, and the ladder rungs are not testing anything.

2 Model

Assume a correctly specified 2PL item response model. Item i has discrimination $a_i > 0$ and difficulty b_i:

$$p_i(\theta) = \sigma\big(a_i(\theta - b_i)\big), \qquad \sigma(x) = \frac{1}{1 + e^{-x}}. \tag{1}$$

3 Step 1: Fisher information from the acquired context

The Fisher information that a single response on item i carries about θ is

$$I_i(\theta) = a_i^2 \, p_i(\theta)\big(1 - p_i(\theta)\big). \tag{2}$$

For a context C of n acquired items, the total information at the true ability is additive:

$$I_n(\theta) = \sum_{i \in C} a_i^2 \, p_i(\theta)\big(1 - p_i(\theta)\big). \tag{3}$$

This is exactly the quantity already computed as the negative Hessian in the existing MAP solver (`faithful_np.py`, `map_theta`):

```
hessian = -sum(a_C^2 * p_C * (1 - p_C)) - 1 / prior_variance
```

The codebase is already computing $I_n(\theta)$ at every forward pass; it is simply not being read as a bound.

4 Step 2: posterior variance (Laplace approximation)

With a Gaussian prior of variance τ^2 on θ (the code uses $\tau^2 = 4$), the Laplace approximation to the posterior at the MAP estimate $\hat{\theta}$ gives

$$\sigma_\theta^2(n) \;\approx\; \frac{1}{I_n(\hat{\theta}) + 1/\tau^2}. \tag{4}$$

By the Cramér–Rao bound, no estimator — linear, Bayesian, or a trained neural network — can achieve a lower variance than Eq. (4) if the 2PL model is correctly specified and (a_i, b_i) are known. This is the hard floor referred to loosely as "$\sim$ 31 bits of information": the precise statement is that $\sigma_\theta^2(n)$ is bounded below by the reciprocal of total Fisher information, not literally Shannon bits.

5 Step 3: propagating ability uncertainty into item-level Brier risk

For a target item t with parameters (a_t, b_t), the Bayes-optimal predictor under squared loss is the posterior-predictive probability

$$\hat{p}_t = \mathbb{E}_{\theta \sim q(\theta|C)}\big[p_t(\theta)\big]. \tag{5}$$

Its achievable risk decomposes exactly into an *estimation* term and an *irreducible noise* term. For any fixed θ,

$$\mathbb{E}_{y \sim \mathrm{Bernoulli}(p_t(\theta))}\big[(\hat{p}_t - y)^2 \mid \theta\big] = (\hat{p}_t - p_t(\theta))^2 + p_t(\theta)\big(1 - p_t(\theta)\big). \tag{6}$$

Taking the expectation over $\theta \sim q(\theta \mid C)$ and using $\hat{p}_t = \mathbb{E}_\theta[p_t(\theta)]$ for the first term:

$$R(t) \;=\; \underbrace{\mathrm{Var}_{\theta \sim q}\big[p_t(\theta)\big]}_{\text{residual ability uncertainty}} \;+\; \underbrace{\mathbb{E}_{\theta \sim q}\big[p_t(\theta)\big(1 - p_t(\theta)\big)\big]}_{\text{irreducible Bernoulli noise}}. \tag{7}$$

This is the Bayes risk floor for item t: it is what the best possible predictor achieves, given only the information content of the context, with the true item parameters known exactly.

5.1 First-order closed form

Linearizing $p_t(\theta)$ around $\hat{\theta}$, $p_t(\theta) \approx \hat{p}_t + a_t\,\hat{p}_t(1 - \hat{p}_t)(\theta - \hat{\theta})$, and substituting $\sigma_\theta^2(n)$ from Eq. (4) gives a closed-form ceiling that needs no simulation:

$$R(t; n) \;\approx\; \hat{p}_t\big(1 - \hat{p}_t\big) \;+\; a_t^2\,\hat{p}_t^2\big(1 - \hat{p}_t\big)^2\,\sigma_\theta^2(n), \qquad \hat{p}_t = p_t(\hat{\theta}). \tag{8}$$

6 Step 4: the ceiling to compare against

Using the competition's own budget grid and weights,

$$\mathrm{ALC}_{\min} \;=\; \sum_{b \in \{0,1,3,7,15,31\}} w_b \cdot \mathbb{E}_{\mathrm{tasks},\,t}\big[R(t;\, n = b)\big], \tag{9}$$

where the expectation over target items t and tasks is taken either

- **in closed form**, by plugging the corpus's empirical distribution of (a_i, b_i) into Eq. (8) and averaging, or
- **by Monte Carlo**: simulate $\theta_{\mathrm{true}} \sim \mathcal{N}(0, \tau^2)$, draw context and target responses from real or fitted (a_i, b_i), run the *existing* MAP + Hessian solver exactly as in production, evaluate $R(t)$ by quadrature over the resulting $q(\theta)$, and average over many simulated tasks.

7 The check

Compare $\mathrm{ALC}_{\min}$ from Eq. (9) against the measured Beta-Binomial Brier ALC on validation (0.222838, per `uniform-audit.md`).

- If the measured value is within noise of $\mathrm{ALC}_{\min}$, the baseline is already near-optimal given the information content of ≤ 31 binary responses. No architecture — CNP, ANP, GP-residual, or latent — can beat it without violating the Cramér–Rao bound, and ladder rungs 3 through 6 are testing against an unbeatable number rather than against a weak baseline.
- If the measured value is meaningfully above $\mathrm{ALC}_{\min}$, there is real headroom, and the null results documented in `ablation-results.md` and `uniform-audit.md` point to a training or objective problem (undertraining, KL misweighting, detached MAP gradient) rather than to a fundamental information limit.

This is a half-day closed-form-plus-Monte-Carlo computation against code that already exists (`map_theta`), and it should be run before committing further compute to architecture rungs 3–6 of the ablation ladder.